1.

```
[loki@loki ~]$ sudo mn --controller=remote --topo=single,4
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet>
```

2.

```
mininet> h1 ping h4 -c 5
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=1.94 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=0.496 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=0.044 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=0.146 ms
64 bytes from 10.0.0.4: icmp_seq=5 ttl=64 time=0.075 ms

--- 10.0.0.4 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4058ms
rtt min/avg/max/mdev = 0.044/0.539/1.937/0.717 ms
mininet>
```

3.

```
DEBUG:forwarding.l2_learning:Connection [00-00-00-00-00-01 2]
DEBUG:forwarding.l2_learning:installing flow for fe:4f:00:48:fd:2e.2 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for d2:d9:75:7a:cf:76.1 -> fe:4f:00:48:fd:2e.2
DEBUG:forwarding.l2_learning:installing flow for fe:4f:00:48:fd:2e.2 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for b6:42:5e:fb:57:9a.3 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for d2:d9:75:7a:cf:76.1 -> b6:42:5e:fb:57:9a.3
DEBUG:forwarding.l2_learning:installing flow for b6:42:5e:fb:57:9a.3 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for d2:d9:75:7a:cf:76.1 -> 32:98:e9:0e:08:5c.4
DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for fe:4f:00:48:fd:2e.2 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for d2:d9:75:7a:cf:76.1 -> fe:4f:00:48:fd:2e.2
DEBUG:forwarding.l2_learning:installing flow for b6:42:5e:fb:57:9a.3 -> fe:4f:00:48:fd:2e.2
DEBUG:forwarding.l2_learning:installing flow for fe:4f:00:48:fd:2e.2 -> b6:42:5e:fb:57:9a.3
DEBUG:forwarding.l2_learning:installing flow for b6:42:5e:fb:57:9a.3 -> fe:4f:00:48:fd:2e.2
DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> fe:4f:00:48:fd:2e.2
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DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> fe:4f:00:48:fd:2e.2
DEBUG:forwarding.l2_learning:installing flow for b6:42:5e:fb:57:9a.3 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for d2:d9:75:7a:cf:76.1 -> b6:42:5e:fb:57:9a.3
DEBUG:forwarding.l2_learning:installing flow for b6:42:5e:fb:57:9a.3 -> fe:4f:00:48:fd:2e.2
DEBUG:forwarding.l2_learning:installing flow for fe:4f:00:48:fd:2e.2 -> b6:42:5e:fb:57:9a.3
DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> b6:42:5e:fb:57:9a.3
DEBUG:forwarding.l2_learning:installing flow for b6:42:5e:fb:57:9a.3 -> 32:98:e9:0e:08:5c.4
DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> b6:42:5e:fb:57:9a.3
DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for d2:d9:75:7a:cf:76.1 -> 32:98:e9:0e:08:5c.4
DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> fe:4f:00:48:fd:2e.2
DEBUG:forwarding.l2_learning:installing flow for fe:4f:00:48:fd:2e.2 -> 32:98:e9:0e:08:5c.4
DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> b6:42:5e:fb:57:9a.3
DEBUG:forwarding.l2_learning:installing flow for b6:42:5e:fb:57:9a.3 -> 32:98:e9:0e:08:5c.4
DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> fe:4f:00:48:fd:2e.2
DEBUG:forwarding.l2_learning:installing flow for b6:42:5e:fb:57:9a.3 -> fe:4f:00:48:fd:2e.2
DEBUG:forwarding.l2_learning:installing flow for b6:42:5e:fb:57:9a.3 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for 32:98:e9:0e:08:5c.4 -> b6:42:5e:fb:57:9a.3
DEBUG:forwarding.l2_learning:installing flow for fe:4f:00:48:fd:2e.2 -> d2:d9:75:7a:cf:76.1
DEBUG:forwarding.l2_learning:installing flow for d2:d9:75:7a:cf:76.1 -> 32:98:e9:0e:08:5c.4
DEBUG:forwarding.l2_learning:installing flow for fe:4f:00:48:fd:2e.2 -> 32:98:e9:0e:08:5c.4
DEBUG:forwarding.l2_learning:installing flow for fe:4f:00:48:fd:2e.2 -> b6:42:5e:fb:57:9a.3
DEBUG:forwarding.l2_learning:installing flow for d2:d9:75:7a:cf:76.1 -> b6:42:5e:fb:57:9a.3
DEBUG:forwarding.l2_learning:installing flow for b6:42:5e:fb:57:9a.3 -> 32:98:e9:0e:08:5c.4
DEBUG:forwarding.l2_learning:installing flow for d2:d9:75:7a:cf:76.1 -> fe:4f:00:48:fd:2e.2
DEBUG:openflow.of_01:1 connection aborted
```

4. Latency

```
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=3.34 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=0.493 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=0.114 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=0.304 ms
64 bytes from 10.0.0.4: icmp_seq=5 ttl=64 time=0.131 ms
64 bytes from 10.0.0.4: icmp_seq=6 ttl=64 time=0.070 ms
64 bytes from 10.0.0.4: icmp_seq=7 ttl=64 time=0.042 ms
64 bytes from 10.0.0.4: icmp_seq=8 ttl=64 time=0.088 ms
64 bytes from 10.0.0.4: icmp_seq=9 ttl=64 time=0.074 ms
64 bytes from 10.0.0.4: icmp_seq=10 ttl=64 time=0.100 ms
```

5. Dumpflow

```
*** Results: 0% dropped (12/12 received)
mininet> sh ovs-ofctl dump-flows s1
 cookie=0x0, duration=2.408s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priori
y=65535,arp,in_port="s1-eth2",vlan_tci=0x0000,dl_src=7e:96:2e:ff:a6:52,dl_dst=de:f9:72:bb:43:f3,arp_spa=
0.0.0.2,arp_tpa=10.0.0.1,arp_op=2 actions=output:"s1-eth1"
 cookie=0x0, duration=2.402s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priori
y=65535,arp,in_port="s1-eth3",vlan_tci=0x0000,dl_src=1a:72:3f:d8:9c:c8,dl_dst=de:f9:72:bb:43:f3,arp_spa=
0.0.0.3,arp_tpa=10.0.0.1,arp_op=2 actions=output:"s1-eth1"
 cookie=0x0, duration=2.396s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priori
y=65535,arp,in_port="s1-eth4",vlan_tci=0x0000,dl_src=b2:c2:b0:cc:d2:80,dl_dst=de:f9:72:bb:43:f3,arp_spa=
0.0.0.4,arp_tpa=10.0.0.1,arp_op=2 actions=output:"s1-eth1"
 cookie=0x0, duration=2.388s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priori
y=65535,arp,in_port="s1-eth3",vlan_tci=0x0000,dl_src=1a:72:3f:d8:9c:c8,dl_dst=7e:96:2e:ff:a6:52,arp_spa=
0.0.0.3,arp_tpa=10.0.0.2,arp_op=2 actions=output:"s1-eth2"
 cookie=0x0, duration=2.385s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priori
y=65535,arp,in_port="s1-eth4",vlan_tci=0x0000,dl_src=b2:c2:b0:cc:d2:80,dl_dst=7e:96:2e:ff:a6:52,arp_spa=
0.0.0.4,arp_tpa=10.0.0.2,arp_op=2 actions=output:"s1-eth2"
 cookie=0x0, duration=2.372s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priori
y=65535,arp,in_port="s1-eth4",vlan_tci=0x0000,dl_src=b2:c2:b0:cc:d2:80,dl_dst=1a:72:3f:d8:9c:c8,arp_spa=
0.0.0.4,arp_tpa=10.0.0.3,arp_op=2 actions=output:"s1-eth3"
 cookie=0x0, duration=2.407s, table=0, n_packets=2, n_bytes=196, idle_timeout=10, hard_timeout=30, prior
ty=65535,icmp,in_port="s1-eth1",vlan_tci=0x0000,dl_src=de:f9:72:bb:43:f3,dl_dst=7e:96:2e:ff:a6:52,nw_src=
10.0.0.1,nw_dst=10.0.0.2,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth2"
 cookie=0x0, duration=2.406s, table=0, n_packets=1, n_bytes=98, idle_timeout=10, hard_timeout=30, priori
y=65535,icmp,in_port="s1-eth2",vlan_tci=0x0000,dl_src=7e:96:2e:ff:a6:52,dl_dst=de:f9:72:bb:43:f3,nw_src=
0.0.0.2,nw_dst=10.0.0.1,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth1"
 cookie=0x0, duration=2.401s, table=0, n_packets=1, n_bytes=98, idle_timeout=10, hard_timeout=30, priori
y=65535,icmp,in_port="s1-eth1",vlan_tci=0x0000,dl_src=de:f9:72:bb:43:f3,dl_dst=1a:72:3f:d8:9c:c8,nw_src=
0.0.0.1,nw_dst=10.0.0.3,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth3"
 cookie=0x0, duration=2.400s, table=0, n_packets=1, n_bytes=98, idle_timeout=10, hard_timeout=30, priori
y=65535,icmp,in_port="s1-eth3",vlan_tci=0x0000,dl_src=1a:72:3f:d8:9c:c8,dl_dst=de:f9:72:bb:43:f3,nw_src=
0.0.0.3,nw_dst=10.0.0.1,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth1"
 cookie=0x0, duration=2.396s, table=0, n_packets=1, n_bytes=98, idle_timeout=10, hard_timeout=30, priori
y=65535,icmp,in_port="s1-eth1",vlan_tci=0x0000,dl_src=de:f9:72:bb:43:f3,dl_dst=b2:c2:b0:cc:d2:80,nw_src=
0.0.0.1,nw_dst=10.0.0.4,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth4"
 cookie=0x0, duration=2.395s, table=0, n_packets=1, n_bytes=98, idle_timeout=10, hard_timeout=30, priori
y=65535,icmp,in_port="s1-eth4",vlan_tci=0x0000,dl_src=b2:c2:b0:cc:d2:80,dl_dst=de:f9:72:bb:43:f3,nw_src=
0.0.0.4,nw_dst=10.0.0.1,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth1"
 cookie=0x0, duration=2.392s, table=0, n_packets=1, n_bytes=98, idle_timeout=10, hard_timeout=30, priori
y=65535,icmp,in_port="s1-eth2",vlan_tci=0x0000,dl_src=7e:96:2e:ff:a6:52,dl_dst=de:f9:72:bb:43:f3,nw_src=
0.0.0.2,nw_dst=10.0.0.1,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth1"
 cookie=0x0, duration=2.391s, table=0, n_packets=1, n_bytes=98, idle_timeout=10, hard_timeout=30, priori
y=65535,icmp,in_port="s1-eth1",vlan_tci=0x0000,dl_src=de:f9:72:bb:43:f3,dl_dst=7e:96:2e:ff:a6:52,nw_src=
0.0.0.1,nw_dst=10.0.0.2,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth2"
 cookie=0x0, duration=2.388s, table=0, n_packets=1, n_bytes=98, idle_timeout=10, hard_timeout=30, priori
y=65535,icmp,in_port="s1-eth2",vlan_tci=0x0000,dl_src=7e:96:2e:ff:a6:52,dl_dst=1a:72:3f:d8:9c:c8,nw_src=
0.0.0.2,nw_dst=10.0.0.3,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth3"
 cookie=0x0, duration=2.388s, table=0, n_packets=1, n_bytes=98, idle_timeout=10, hard_timeout=30, priori
y=65535,icmp,in_port="s1-eth3",vlan_tci=0x0000,dl_src=1a:72:3f:d8:9c:c8,dl_dst=7e:96:2e:ff:a6:52,nw_src=
0.0.0.3,nw_dst=10.0.0.2,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth2"
 cookie=0x0, duration=2.385s, table=0, n_packets=1, n_bytes=98, idle_timeout=10, hard_timeout=30, priori
```